

Determination of National Income

1. Different concepts of National income: Formulas:

◆ GDP Deflator:

$$\text{GDP Deflator} = \frac{\text{Current Price GDP}}{\text{Base Price GDP}} \times 100$$

◆ Gross National Product (GNP):

$$\text{GNP}_{\text{MP}} = \text{GDP}_{\text{MP}} + \text{Net Factor Income from Abroad}$$

◆ Net National Product at Market Prices (NNP_{MP}):

$$\text{NNP}_{\text{MP}} = \text{GNP}_{\text{MP}} - \text{Depreciation}$$

◆ Net Domestic Product at Factor Cost (NDP_{FC}):

$$\text{NDP}_{\text{FC}} = \text{NDP}_{\text{MP}} - \text{Net Indirect Taxes}$$

◆ Per Capita Income:

$$\text{PCI} = \frac{\text{NDP}_{\text{FC}}}{\text{Population}}$$

◆ Disposable Income (DI):

$$\text{DI} = \text{PI} - \text{Personal Income Taxes} - \text{Non tax payments}$$

2. Basic Concepts and Functions:

◆ The consumption function, proposed by Keynes:

$$C = a + bY$$

◆ Average propensity to consume (APC):

$$\text{APC} = \frac{C}{Y}$$

◆ Marginal propensity to consume (MPC):

$$\text{MPC} = \frac{\Delta C}{\Delta Y}$$

◆ Average propensity to save (APS):

$$\text{APS} = \frac{S}{Y}$$

◆ Marginal propensity to save (MPS):

$$\text{MPS} = \frac{\Delta S}{\Delta Y}$$

$$1. \text{APC} + \text{APS} = 1$$

$$2. \text{MPC} + \text{MPS} = 1$$

◆ Investment multiplier:

$$k = \frac{1}{1 - \text{MPC}} \text{ or } \frac{1}{\text{MPS}}$$

3. Personal Income:

◆ $\text{PI} = \text{NI} - \text{Undistributed profits} - \text{Net interest payments made by households} - \text{Corporate Tax} + \text{Transfer Payments to households from firms and the government.}$

4. Components of Aggregate Demand or Equilibrium Level of Income:

◆ In two sector economy: $Y = C + I$

◆ In three sector economy: $Y = C + I + G$

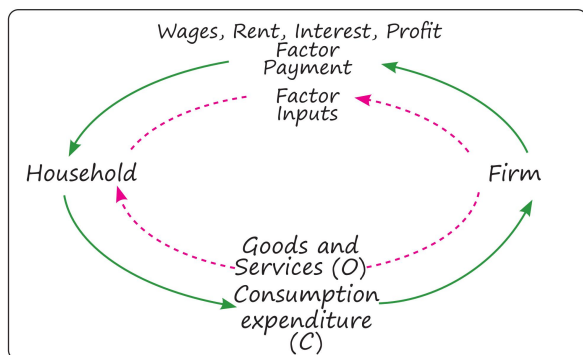
◆ In four sector economy: $Y = C + I + G + (X - M)$

5. Private Income:

Private Income = Factor income from net domestic product accruing to the private sector + Net factor income from abroad + National debt interest + Current transfers from government + Other net transfers from the rest of the world.

6. The Circular Flow of Income: Two-Sector Economy:

A firm produces goods/services with the help of factor services. The flow of factor income, such as rent, flows from firms to households.



Factor Payments = Household Income = Household Expenditure = Total Receipts of Firms = Value of Output

7. Deflationary Gap Vs. Inflationary Gap:

- ◆ **Deflationary Gap:** $AD < \text{Full Employment Output}$, or called Deficient Demand.
- ◆ **Inflationary Gap:** $AD > \text{Full Employment Output}$, or called Excess Demand.

8. Measurement of National Income in India:

- ◆ **Value Added Method/Product Method/Industrial Origin Method:**

$$NVA \text{ at FC} = GVA_{MP} - Dep - NIT$$

◆ Income Method/Factor Payment Method/Distributed Share Method:

NNP_{FC} = Compensation of employees + Operating surplus (rent + interest + profit + royalty) + Mixed income of self-employed + Net factor income from abroad (NFIA).

◆ Expenditure Method/Income Disposable Method:

$$GDP_{MP} = C + I + G + NX$$

9. LEAKAGE and INJECTIONS in different sectors:

Model	Leakage	Injections
Two Sector	Savings	Investment
Three Sector	Savings + Taxes	Investment + Govt. Exp.
Four Sector Model	Savings + Taxes + Imports	Investment + Govt. Exp. + Exports